Pre-registration for “Nestedness in phonological inventories of world's languages’’

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Contributions

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I confirm the accuracy of this Contributions statement. OM

1. Introduction

The notion of nestedness corresponds to a phenomenon whereby assemblages contain both common and rare elements, but tend not to possess exclusively rare elements. Initially introduced in ecology (Ulrich et al., 2009), it is based on the observation that ecosystems rich in rare species also contain common species, while environments with low biodiversity contain mostly common species. For instance, the set of species in an island usually corresponds to a subset of species from the nearby mainland composed mostly from common species. In practice, nestedness is usually modelled using bipartite graphs.

Nestedness is frequently found in domains other than ecology. For example, individuals that know rare facts are also shown to possess extensive general knowledge, but not vice-versa (Dubourg et al. in press). Additionally, Morin and Sobchuk (2021) have shown that film collections follow a similar pattern: films in restricted collections are also present in larger collections, but the inverse is rarely true. However, they also found certain exceptions, such as the *film noir* movies, which remain rare even in large collections, while being present in small specialised collections, demonstrating that the notion of nestedness is not absolute.

In the same way that nestedness appears in ecological and cultural domains, it is also likely to manifest itself in linguistics. It is an open question whether the phonological inventories of languages obey similar dynamics as other cultural inventories, like the film inventories or factual knowledge. **Is it the case that phonological inventories systematically contain rare phonemes only if they also contain more cross-linguistically common phonemes?** The phenomenon of *markedness* provides a possible cue: if a language has a marked, usually less frequent, phoneme, it will also include its unmarked, usually more frequent, counterpart (Haspelmath, 2006). This is consistent with the view that large phoneme inventories would contain both "marked" (infrequent) and "unmarked" (frequent) phonemes, but smaller inventories only contain "unmarked" (i.e. frequent) phonemes.

Other approaches explain the distribution of phonemes across languages: dispersion theory (Liljencrants and Lindblom, 1972; Flemming, 2001; Vaux and Samuels, 2015), and the optimisation of perceptual and articulatory dimensions, in particular with the notions of global and local economy and symmetry (Dunbar and Dupoux, 2016). These approaches also appear, *prima facie*, consistent with nestedness. The principle of optimal dispersal posits that, given certain phonemes, other phonemes are much more likely to occur, because they are optimally distant (we think that symmetry and optimal dispersal are two ultimately convergent ways of expressing this principle). This suggests that, when a phoneme inventory gets augmented, it gains first the phonemes that are optimally distinctive, and only secondarily those that are not. The result should be that inside any language family, there will be a core set of maximally distinctive phonemes, present in most languages including those with very small phoneme inventories. This would be consistent with nestedness.

However, these system-internal principles are not enough to explain the organisation of phonological inventories. Nikolaev (2019) has shown that contacts between neighbouring languages strongly influence their phonology; geographical proximity being a better predictor of phoneme inventory composition than affiliation to a linguistic family. A major factor is phonological borrowing, defined by Moran, Eisen, Nikolaev and Grossman (2024) as 'the acquisition of new sounds by a language through the integration of borrowed words'. This interest in phonetic borrowings has led to the creation of databases such as SEGBO (Grossman, Eisen, Nikolaev and Moran, 2020), making it possible to quantify these exchanges on a large scale. Their work shows that consonants, intermediate frequency segments and labiodental segments are the most likely to be borrowed across the languages of the world. While this body of work does not directly predict nestedness in phonological inventories, neither does it contradict this possibility, since their work indicates that very common phonemes are rarely borrowed.

Hypotheses such as markedness and optimal dispersion lead us to hypothesize that phonological inventories should be nested. Moreover, the existence of large-scale phonological databases like PHOIBLE greatly facilitates testing of such hypotheses, by offering standardized phonological inventories from a large sample of languages from diverse families.

Nestedness will be operationalised in the way that ecologists operationalise it, using two distinct measurements; NODF (Nestedness metric based on Overlap and Decreasing Fill) (Almeida-Neto et al, 2008) and Nestedness Temperature Calculator (NTC) (Atmar & Patterson, 1993). We will rely on both of these measures to assess the extent to which the phonemes present in relatively small phoneme inventories tend to be a subset of the phonemes present in larger phoneme inventories. A phoneme inventory that refutes our hypothesis is one that has few phonemes, which are nonetheless absent from the larger phoneme inventories. Nestedness is a property not of individual phonological inventories but of sets thereof. We will measure the nestedness as a property of language families, and assess it against several credible baselines.

If we find that a substantial number of language families are nested, this could be due to cultural transmission inside these families, or to more general constraints that affect phonology regardless of phylogeny. In order to test the latter interpretation, we will once again measure nestedness, this time not on language families but on the whole set of languages, disregarding their family affiliations.

2. Data & methods

2.1. Data sources and pre-processing

To obtain the necessary phoneme data, this study uses the PHOIBLE database (Moran, McCloy, & Wright, 2014); a cross-linguistic repository of phonological inventories compiled from source documents and third-party databases. Version 2.0, released in 2019, includes 3,020 inventories, and contains 3,183 unique segment types from 2,186 distinct languages.

The data is available online in Cross-Linguistic Data Formats (CLDF) (Steven Moran, & Daniel McCloy, 2019). For each language, the PHOIBLE database provides at least one phonemic inventory, listing all the corresponding phonemes and the language family to which the language belongs. In this study, we will only rely on the following information: the name of the language, its linguistic family, the name of the inventory and the list of the corresponding phonemes.

The PHOIBLE developers describe phonemes in ways that were intended to be descriptively adequate across languages. Each combination of phonological features is encoded in PHOIBLE by a single symbol of the International Phonetic Alphabet (IPA). If two phonemes differ in their IPA (Unicode) representation, they necessarily also differ in their representation in terms of phonological features. Additional material from the PHOIBLE team can be used to translate those symbols into featural information, but we will only rely on the symbolic representations, as they are equivalent.

In concrete terms, the matrix is organised as follows: the columns (x axis) represent all the phonemes appearing at least once in the inventories of each language in the family, while the rows (y axis) correspond to the languages belonging to this family. Each cell of the matrix indicates the presence (1) or absence (0) of a phoneme in the phoneme inventory of the language concerned. Such a binary matrix is identical to a bipartite graph, although not represented with edges and vertices.

2.2. Inclusion criteria

To ensure the reliability of the analyses, several criteria were applied to filter the data before constructing the phonemic matrices of the language families. We started with 176 families covering 2,186 languages.

1) Language family filtering

Families with strictly fewer than 4 languages were excluded from the study. A matrix of 3 rows or less would make the calculation of nestedness unreliable, as the results are likely to be random. After this selection, 72 language families remained covering 2,020 languages in total.

2) Exclusion of invalid families

Languages without any family attribution were removed, as they do not correspond to any real linguistic classification. After this selection, 70 language families remained with 1,938 languages in total. (The two excluded families at this step are the empty family (for NA family languages), and another dummy family called "bookkeeping".) All families are listed in the appendix.

3) Managing multiple inventories per language

A language may have several phonological inventories from different sources, which may vary in terms of the number of phonemes. On average, each language has 1.4 inventories. To guarantee a single representation per language is used in each matrix (a single row per language), a single inventory will be selected for each language, in the following way:

- When a language has exactly 2 inventories attached to it in PHOIBLE, the smaller of the two, i.e. the one with the lesser number of phonemes, will be picked. Ties will be broken randomly.
- When a language has > 2 inventories attached to it in PHOIBLE, the inventory chosen will be the one with the highest degree of overlap (defined as including the same phonemes) as the others. Ties will be broken by choosing the smallest inventory. If after that there are still ties, they will be broken randomly.

This will ensure that we will rely on the inventory providing the highest expert "agreement" in terms of phonological inventory composition for a given language. We will then build a binary matrix of absences/presences of individual phonemes across all of the chosen inventories for each linguistic family.

4) Segment types taken into account

All segment types (consonants, vowels and tones) have been retained. Some inventories report phonemes as 'marginal', i.e. used only in loanwords. However, these marginal segments represent less than 2% of all phonemes. We have therefore chosen to keep them in the analysis.

In addition to our individual analysis of the 70 language families, we will also test the nestedness of the entire dataset considered as one single matrix. This will allow us to assess whether global preferences in the constitution of phoneme inventories (either due to universal cognitive preferences or to deep cultural commonalities) may result in nestedness independently of any family-specific signal.

2.3. Data analysis

To quantify the degree of nestedness, we use two measures derived from ecology: the Nestedness Temperature Calculator (NTC) and the Nestedness metric based on Overlap and Decreasing Fill (NODF), both available in the vegan package in R (Oksanen et al., 2025).

The NTC, proposed by Atmar & Patterson (1993) (see also a review by Rodríguez-Gironés & Santamaria, 2006) measures the degree of randomness in a matrix by comparing its organisation with a theoretically perfect order. It is based on the concept of surprisal, which quantifies the unpredictability of presence and absence. A high temperature indicates high variability and low nestedness, while a low temperature reflects a more systematic, non-random, structure. This measure is standardized on a scale from 0, which corresponds to perfect nestedness to 100, corresponding to lowest nestedness.

The NODF measure, introduced by Almeida-Neto et al (2008), measures the nestedness of a matrix by evaluating two criteria: the progressive decrease in marginal totals, where the biggest inventories must contain more 1s than the least rich, and pair overlap, which checks whether the phonemes of the least rich languages are systematically a subset of those of the richest languages. This measure is also normalized between 0 (maximum disorder) and 100 (perfect nestedness).

Once we will compute these measures for every linguistic family in our sample, we will compare these with a random baseline, to determine whether the observed degree of nestedness is not due to chance. To do this, we will generate simulated matrices using two null models: r00 and c0. The r00 baseline only preserves the total number of presences (fills) without preserving the size of the inventories (rows) or the frequency of the phonemes (columns). The c0 baseline is a null model that fixes phoneme frequencies (columns) while randomizing the size of the inventories (rows), see Jonsson (2001).

The use of a more conservative c0 baseline is warranted by an analysis based on the Gini coefficients, which showed that inventory sizes are relatively homogeneous in each language family, whereas phoneme frequency is much more variable (see Figure 1). This inequality suggests that it is preferable to fix the prevalence of phonemes rather than the size of inventories during permutations, since the dataset appears to be structured overwhelmingly by variance in phoneme frequency. A high degree of inequality in phoneme frequency reflects an important underlying structure that could be destroyed if left uncontrolled for. c0 is a highly conservative baseline, since high variance in one of the matrix's dimensions will result, by itself, in greater nestedness, bringing the c0 baseline closer to the level of nestedness of the real data. This could be a driver of differences in nestedness between families, since the inequalities in phoneme frequencies are quite variable across families.

If we find that a language family is nested under both baselines, c0 and r00, we will interpret this as meaning that the phoneme inventories in that family are nested, and that this is not explained by inequality in phoneme prevalence alone. If we find that a language family is nested under r00 only, we will interpret this as showing that the phoneme inventories in that family are nested, nestedness being due to the fact that some phonemes are more prevalent than others. We do not expect any substantial number of families to show nestedness under c0 but not r00, c0 being intrinsically more conservative. If a family is not significantly nested against either baseline, this can be interpreted in one of two ways: either the data is insufficiently abundant to detect nestedness, or nestedness is absent. The former interpretation would be strengthened if we find that we are more likely to detect nestedness in bigger families (with more languages).

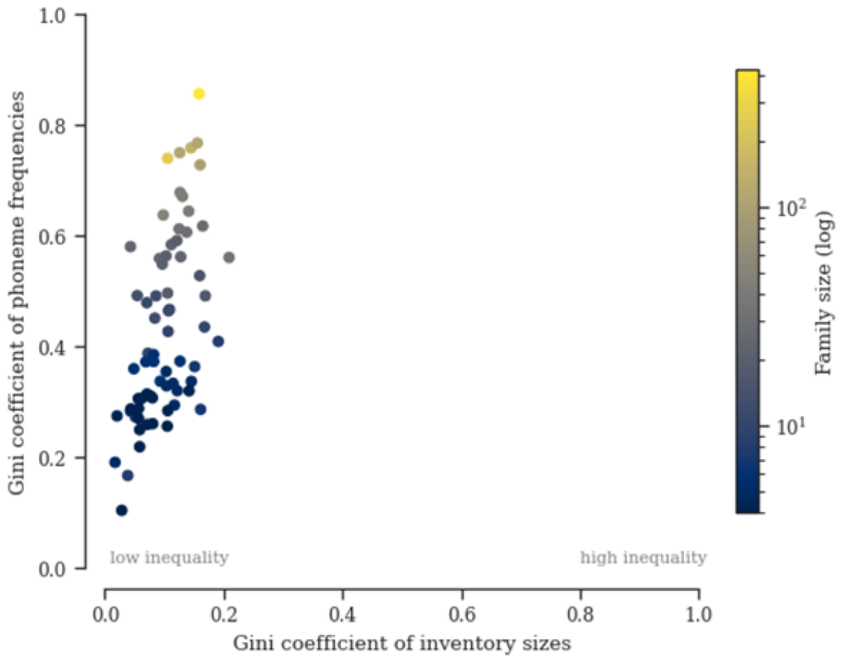

Figure 1 : Gini coefficient of inventory sizes and phoneme frequencies

For each of the 70 families in our sample, 10,000 random matrices will be generated with the r00 baseline and another 10,000 with the c0 baseline, both for NODF and for temperature. This will result in 40,000 simulated values in total.

The simulations will be carried out using the oecosimu function in the vegan package in R, which generates a two-sided p-value indicating how many of the simulated matrices display a higher or lower NODF score or a higher or lower temperature than the empirical matrix. A high proportion of such occurrences suggests that the structure observed in the empirical matrices is not due to chance. We will set the significance level at 0.005, ensuring that only statistically robust differences are taken into account.

We predict that a substantial number of language families will display significant degrees of nestedness against r00, on both measures (NTC and NODF). Additionally, we will test for nestedness against the c0 baseline. We have committed to an interpretation of the results, see above. This number should be greater than the number of families displaying the opposite behaviour (the real data being significantly less nested than the baselines).

2.4. Dataset structure

The dataset that we will use to test our hypothesis will contain information on the simulations described above for each linguistic family. It will be organised by type of nestedness metric, with each row representing a specific observation. The dataset will include the following variables:

1) Family - The language family to which the data point belongs.

2) Measure - The specific measure analysed (NODF or Temperature).

3) Type - Indicates whether the value comes from a simulated or empirical dataset.

4) Value - The numerical result of the measurement.

5) p_value - The p-value associated with the statistical significance of the measurement (two-sided).

6) significant - A Boolean indicator (True/False) indicating whether the result is statistically significant.

7) Simulated_Mean - The average nestedness score of the simulations

8) Significant_Side - The side for which the p-value is significant (nested, antinested or none if it is not significant)

9) Sig_cat - The number of significant results for a family in total according to the two baselines (one, two or none)

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Appendix: List of language families included in the study

Family name	Number of languages
Atlantic-Congo	423
Pama-Nyungan	253
Indo-European	149
Austronesian	114

Afro-Asiatic	108
Sino-Tibetan	100
Tupian	50
Arawakan	43
Mande	43
Austroasiatic	40
Dravidian	33
Uralic	32
Nuclear Trans New Guinea	30
Nilotic	28
Cariban	24
Pano-Tacanan	24
Quechuan	23
Central Sudanic	22
Turkic	21
Nuclear-Macro-Je	20
Tai-Kadai	19
Tucanoan	19

Otomanguean	17
Uto-Aztecan	15
Gunwinyguan	14
Nakh-Daghestanian	13
Mayan	12
Mongolic	12
Chibchan	11
Salishan	11
Athabaskan-Eyak-Tlingit	10
Nyulnyulan	10
Ta-Ne-Omotic	10
Mirndi	8
Worrorran	8
Kadugli-Krongo	7
Sepik	7
Algic	6
Chocoan	6
Heibanic	6

Ijoid	6
Mixe-Zoque	6
Nuclear Torricelli	6
Western Daly	6
Arawan	5
Eskimo-Aleut	5
Guaicuruan	5
Iwaidjan Proper	5
Kiowa-Tanoan	5
Nambiquaran	5
Narrow Talodi	5
Surmic	5
Tungusic	5
Barbacoan	4
Chicham	4
Chukotko-Kamchatkan	4
Guahiboan	4
Huitotoan	4

Iroquoian	4
Kartvelian	4
Koman	4
Mangarrayi-Maran	4
Maningrida	4
Matacoan	4
Muskogean	4
Nadahup	4
Saharan	4
Songhay	4
Yanomamic	4
Zaparoan	4